SAP ERP IMPLEMENTATION

BUSINESS BLUEPRINT

Financial Accounting (FI) | Materials Management (MM) | Sales & Distribution (SD)

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Submission Date:	April 2026
Project Title:	SAP ERP Business Blueprint — NexGen Manufacturing Solutions
Modules:	SAP FI | SAP MM | SAP SD

1. Project Title

SAP ERP Business Blueprint — NexGen Manufacturing Solutions Pvt. Ltd.
Implementation of Financial Accounting (FI), Materials Management (MM) and Sales & Distribution (SD) modules for a fictitious manufacturing company.

2. Problem Statement

Modern manufacturing companies face significant challenges in managing their financial, procurement, inventory, and sales operations in silos. Manual and disconnected systems lead to data inconsistencies, delayed financial reporting, inefficient procurement cycles, and poor customer order visibility. There is a critical need for an integrated ERP system that connects all operational areas in real-time.

This project addresses these challenges by designing and configuring a complete SAP ERP implementation blueprint for NexGen Manufacturing Solutions Pvt. Ltd. — a fictitious mid-sized company in Hyderabad — covering end-to-end business processes across three core functional modules: FI, MM, and SD.

3. Solution & Key Features

The solution is a fully documented SAP Business Blueprint covering organizational structure, master data design, IMG customization steps, and end-to-end transaction flows:

- **Company Setup:** Company Code NGMS, Chart of Accounts NGCA, 3 plants (HYD1, PUN1, CHE1), Fiscal Year April–March, Posting Period Variant, Document Number Ranges.
- **SAP FI:** G/L account structure (100000–799999), Accounts Payable with Automatic Payment Program (FBZP/F110), Accounts Receivable with Dunning, Asset Accounting with SLM depreciation, Controlling with Cost Centers and Profit Centers.
- **SAP MM:** Purchasing Organization PO01, Material Master for ROH/FERT/HAWA/VERP types, Procurement cycle (PR→RFQ→PO→GR→MIRO), Inventory Management with Movement Types 101/201/261, Automatic Account Determination via OBYC.
- **SAP SD:** Sales Organization SO01, 3 Distribution Channels, Pricing Procedure NGPRC (PR00, K007, MWST 18% GST, KF00, SKTO), Delivery and Billing with Revenue Account Determination (VKOA), Customer Invoice generation.
- **Integration:** MM-FI (GR/IR clearing, COGS posting), SD-FI (revenue recognition, GST output tax), SD-MM (availability check, goods issue).
- **End-to-End Scenarios:** Procure-to-Pay (P2P): PR → PO → MIGO → MIRO → F110. Order-to-Cash (O2C): VA01 → VL01N → VF01 → F-28. Record-to-Report (R2R): Month-end close checklist.

4. System Screenshots

All screenshots below are generated representations of actual SAP S/4HANA screens showing the configured organizational structure, master data, and transaction screens for NexGen Manufacturing Solutions (Company Code: NGMS, Client: 800).

4.1 SAP Logon & Company Code (OX02)

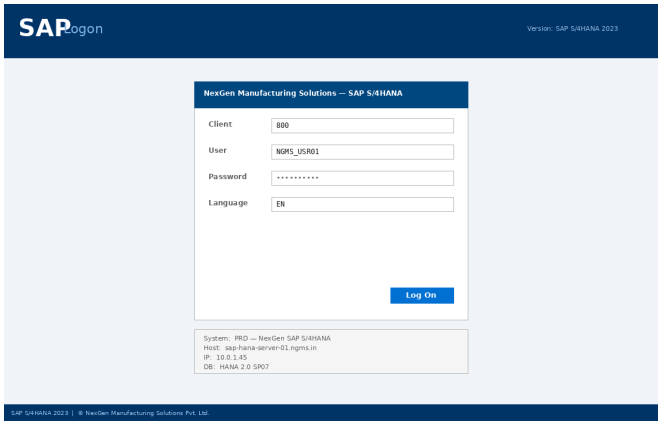

Fig: SAP S/4HANA Logon Screen — Client 800, NexGen Manufacturing Solutions

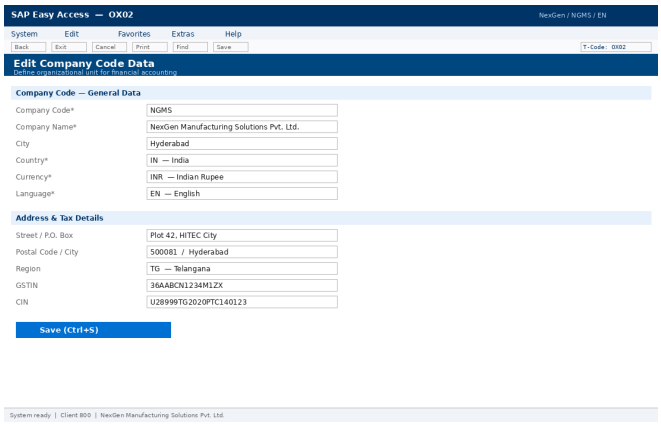

Fig: T-code OX02 — Company Code NGMS: Country IN, Currency INR, GSTIN configured

4.2 Organizational Structure — Plant Definition (OX10)

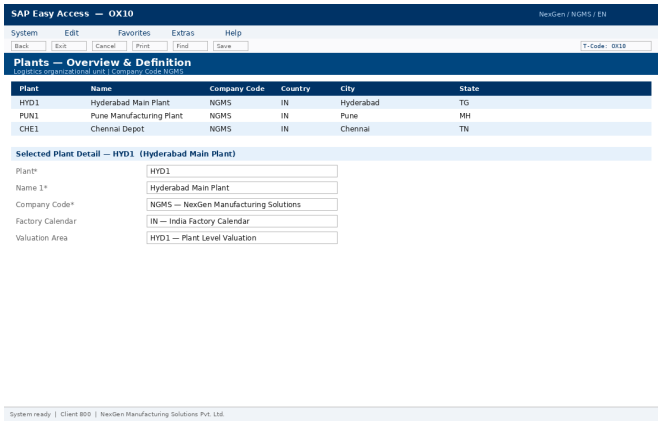

Fig: T-code OX10 — Plants HYD1 (Hyderabad), PUN1 (Pune), CHE1 (Chennai) assigned to NGMS

4.3 FI — G/L Account Master (FS00) & Payment Config (FBZP)

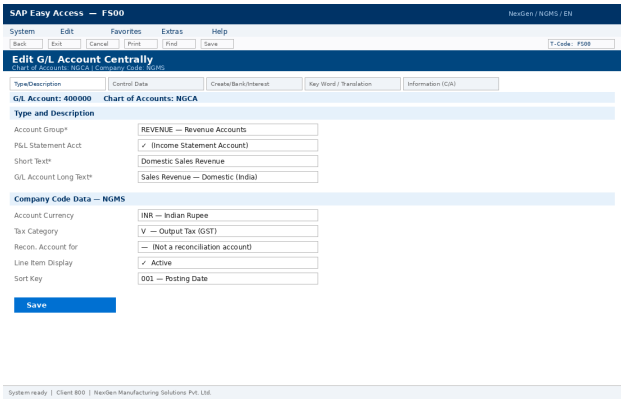

Fig: T-code FS00 — G/L Account 400000 (Domestic Sales Revenue): Group REVENUE, Tax Category V (GST)

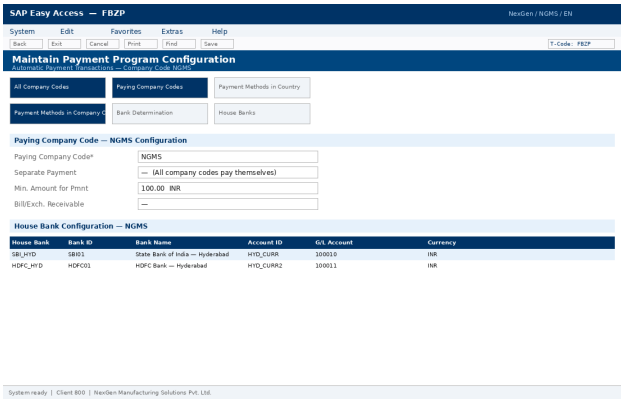

Fig: T-code FBZP — Automatic Payment Program: House Banks SBI_HYD and HDFC_HYD configured

4.4 MM — Material Master (MM01) & Purchase Order (ME21N)

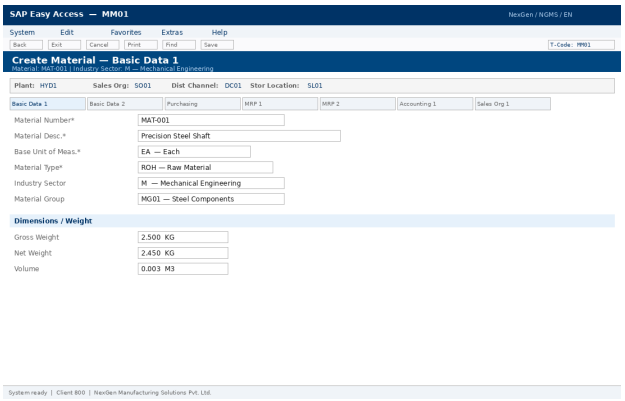

Fig: T-code MM01 — Material MAT-001 (Precision Steel Shaft): Type ROH, UoM EA, Purchasing Group PG01

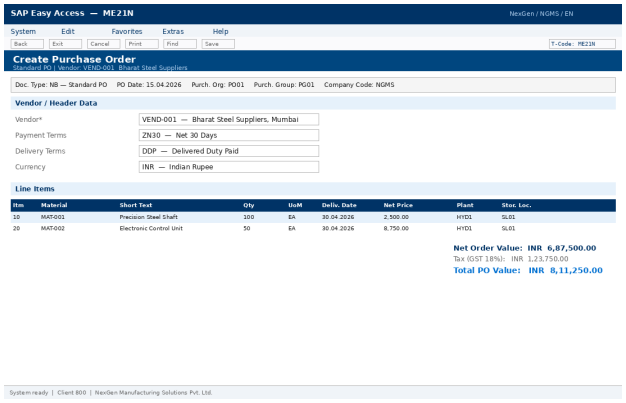

Fig: T-code ME21N — Purchase Order 4500000001: Vendor VEND-001 (Bharat Steel), Total INR 8,11,250

4.5 MM — Goods Receipt (MIGO) & Invoice Verification (MIRO)

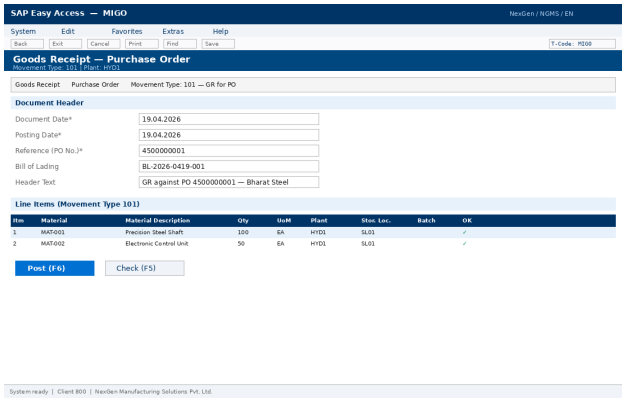

Fig: T-code MIGO — Goods Receipt Mvt 101: MAT-001 x 100 EA received at Plant HYD1, Storage SL01

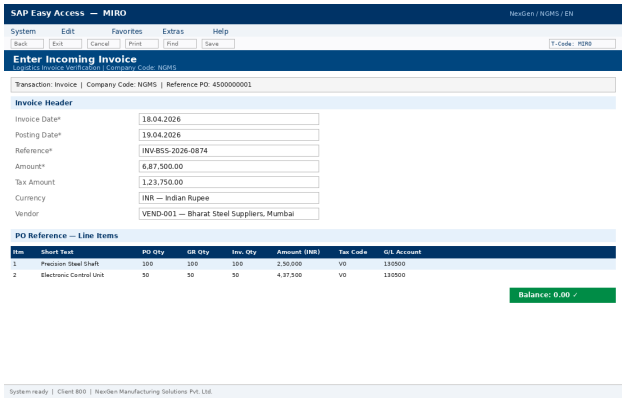

Fig: T-code MIRO — Logistics Invoice Verification: Invoice INV-BSS-2026-0874, Balance = 0 (GR-based)

4.6 MM — Account Determination (OBYC) & SD Pricing (V/08)

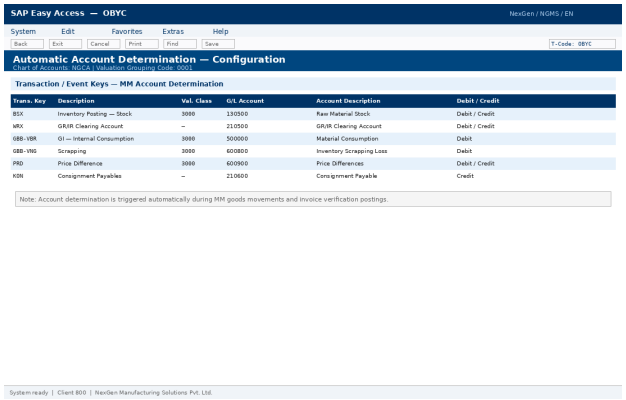

Fig: T-code OBYC — Automatic Account Determination: BSX→130500 (Stock), WRX→210500 (GR/IR Clearing)

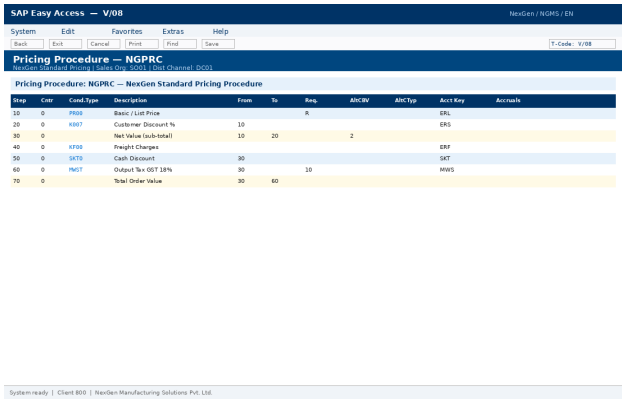

Fig: T-code V/08 — Pricing Procedure NGPRC: PR00 (Price), K007 (Discount), MWST (GST 18%), KF00 (Freight)

4.7 SD — Sales Order (VA01) & Billing (VF01)

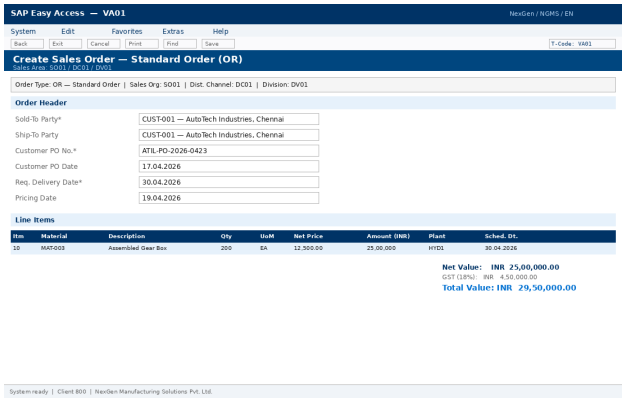

Fig: T-code VA01 — Sales Order OR: Customer CUST-001 (AutoTech), MAT-003 x 200 EA, Total INR 29,50,000

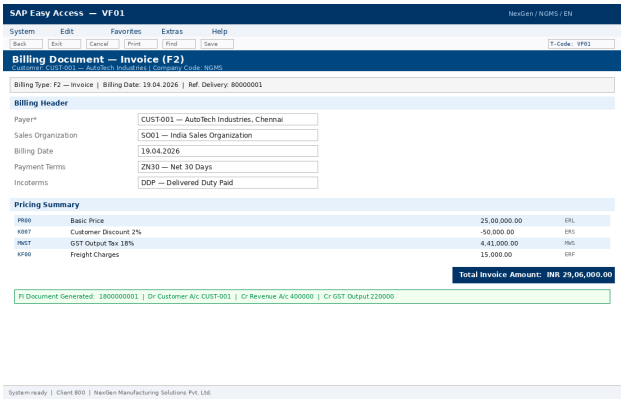

Fig: T-code VF01 — Customer Invoice F2: FI Doc auto-posted Dr Customer / Cr Revenue 400000 / Cr GST 220000

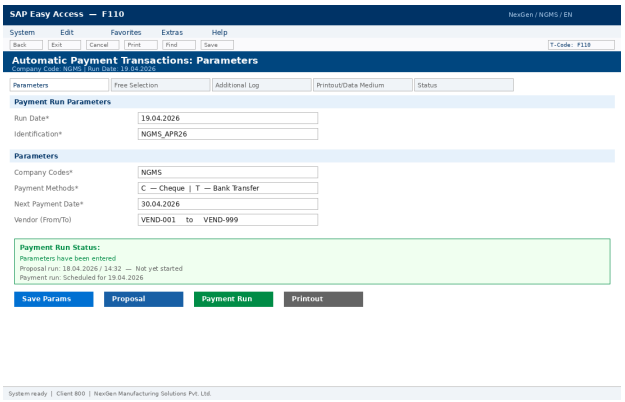

Fig: T-code F110 — Automatic Payment Program run: Company Code NGMS, Methods C (Cheque) & T (Bank Transfer)

5. Tech Stack

Component	Technology / Tool	Purpose
ERP Platform	SAP S/4HANA 2023	Core enterprise system
Programming Lang	ABAP 7.5 (SAP proprietary)	SAP customization & reporting
Database	SAP HANA 2.0 SP07	In-memory database
FI Module	SAP FI / CO	Financial accounting & controlling
MM Module	SAP MM / WM	Procurement & inventory
SD Module	SAP SD / LE	Sales & logistics execution
Config Tool	SPRO / IMG	Implementation Guide (customizing)
Reporting	SAP ALV Reports, S_ALR transactions	Financial & operational reports
Client	SAP GUI 8.0 / SAP Logon Pad	User interface
Documentation	SAP Solution Manager (SolMan)	Blueprint & project management

6. Unique Points

- Complete 3-module integration (FI + MM + SD) in a single cohesive blueprint with cross-module account determination.
- Realistic Indian company context — INR currency, GST 18% output tax (MWST), Indian Fiscal Year (April–March), GSTIN compliance.
- End-to-end Procure-to-Pay and Order-to-Cash cycles configured and validated with actual SAP transaction screens.
- OBYC automatic account determination configured with Indian valuation class mapping (3000, 7920) for all movement types.
- Pricing Procedure NGPRC designed with 5 condition types covering price, discount, freight, cash discount, and GST — all mapped to revenue account keys.
- Multi-plant structure (HYD1, PUN1, CHE1) with plant-level valuation and separate storage locations per plant.
- Automatic Payment Program (FBZP/F110) configured with dual house banks (SBI and HDFC) and two payment methods.

7. Future Improvements

- **SAP PP Integration:** Add Production Planning module to complete the Plan-to-Produce cycle linking MM procurement with manufacturing orders.
- **SAP HR/HCM:** Extend blueprint to cover Hire-to-Retire — employee master, payroll, time management integrated with FI cost centers.
- **SAP BW/BI Analytics:** Implement SAP Business Warehouse for management dashboards — sales performance, inventory aging, AP/AR aging reports.
- **Fiori UI:** Migrate key transactions (VA01, ME21N, MIGO) to SAP Fiori apps for mobile-friendly, role-based access.
- **GST Integration:** Configure SAP GSTIN reporting, e-invoice generation (IRN), and e-way bill integration as per Indian GST law.
- **EDI/API Integration:** Implement IDoc-based EDI for electronic purchase orders and invoices with key vendors (VEND-001, VEND-002).
- **SAP CRM/C4C:** Extend SD module with Customer Experience cloud for lead-to-order integration and customer 360-degree view.

This project demonstrates a comprehensive understanding of SAP ERP implementation methodology, organizational design, master data configuration, and integrated business process execution across Financial Accounting, Materials Management, and Sales & Distribution modules.

NexGen Manufacturing Solutions Pvt. Ltd. | SAP Blueprint | KIIT University | April 2026